

COMPLETE 60 GAMES

IMPLEMENTATION: FRONTEND & BACKEND

CATEGORY 1: THE 30 GOTTMAN GAMES

Game 1: "Bid Radar"

Frontend: React Native component with 60-second timer, bid submission input, real-time partner sync, animated matches display

Backend: Cloud Function for AI analysis of bid matches, real-time Firebase sync, Marcie commentary generation

Database: bidsRadar/{sessionId} with bids array, matches object, timestamps

AI Integration: GPT-4 for semantic similarity analysis between bids

Game 2: "Gentle Start-Up Gauntlet"

Frontend: Timer-based phrase rewrite interface, scoring display, round progression

Backend: Cloud Function analyzing I-statements, structure validation, tone scoring

Database: gentleStartup/{sessionId}/rounds with original phrases, rewrites, scores

AI Integration: NLP analysis for "I feel... about... I need..." structure detection

Game 3: "Love Map Speedrun"

Frontend: Quiz interface with timer, answer input, real-time partner comparison

Backend: Cloud Function generating personalized questions from couple data

Database: loveMap/{sessionId} with questions array, answers objects, accuracy scores

AI Integration: GPT-4 generating personalized questions based on couple history

Game 4: 🛡️ "Antidote Arena"

Frontend: Audio clip player with buzz-in buttons, multiple choice antidote selection

Backend: Cloud Function playing Four Horsemen clips, validating antidote choices

Database: antidoteArena/{sessionId} with clips played, responses, streak counter

AI Integration: ElevenLabs for audio clip generation, GPT-4 for scenario creation

Game 5: 🪞 "Mirror Mode"

Frontend: Video recording interface with prompts, playback comparison, word matching

Backend: Cloud Function processing video responses, tone analysis, word matching

Database: mirrorMode/{sessionId} with video URLs, transcripts, emotional scores

AI Integration: Face/emotion detection, voice sentiment analysis

Game 6: 🕵️ "Dream Decoder"

Frontend: Surface conflict input, clue system, guess interface, reveal animation

Backend: Cloud Function generating underlying dream patterns from conflicts

Database: dreamDecoder/{sessionId} with surface conflicts, underlying dreams, matches

AI Integration: GPT-4 analyzing conflicts to reveal deeper emotional needs

Game 7: 🗣️ "Tone Shift Challenge"

Frontend: Sentence display with 4 tone buttons, recording interface, partner guess input

Backend: Cloud Function analyzing vocal warmth, pitch variation, emotion detection

Database: toneShift/{sessionId} with recordings, tone labels, accuracy scores

AI Integration: Voice emotion analysis, tone classification

Game 8: 🧩 "Ritual Builder"

Frontend: Drag-and-drop ritual components, preview display, partner rating interface

Backend: Cloud Function suggesting ritual improvements, feasibility scoring

Database: ritualBuilder/{sessionId} with rituals built, ratings, improvement suggestions

AI Integration: GPT-4 analyzing ritual components for compatibility and creativity

Game 9: 🎲 "Conflict Dice"

Frontend: Virtual dice roll animation, scenario display, constraint timer, recording interface

Backend: Cloud Function generating conflict scenarios with constraints, analyzing resolutions

Database: conflictDice/{sessionId} with scenarios, recordings, analysis results

AI Integration: GPT-4 generating realistic conflict scenarios, analyzing repair attempts

Game 10: 💌 "Appreciation Auction"

Frontend: Auction interface with bidding coins, anonymous appreciation display, bluff detection

Backend: Cloud Function generating decoy appreciations, running auction logic

Database: appreciationAuction/{sessionId} with appreciations, bids, outcomes

AI Integration: GPT-4 generating plausible decoy appreciations based on couple style

Game 11: 🧘 "Flooding Forecast"

Frontend: Heart rate monitoring via camera, self-report sliders, calm-down game selection

Backend: Cloud Function processing heart rate data, predicting flooding risk

Database: floodingForecast/{sessionId} with biometric data, predictions, outcomes

AI Integration: Heart rate pattern analysis, stress level prediction

Game 12: 🌍 "Love Map Gap Quest"

Frontend: Gap identification display, question crafting interface, rating system

Backend: Cloud Function detecting Love Map gaps, validating curiosity questions

Database: loveMapGaps/{sessionId} with identified gaps, questions, ratings

AI Integration: GPT-4 analyzing couple data for knowledge gaps, evaluating questions

Game 13: 🎨 "Shared Meaning Mural"

Frontend: Collaborative digital canvas, symbol library, real-time sync between devices

Backend: Cloud Function detecting alignment in artistic choices, analyzing symbolic overlap

Database: sharedMeaning/{sessionId} with mural data, alignment scores, analysis

AI Integration: Computer vision analyzing mural for shared symbols and themes

Game 14: 🗣️ "Text Tone Translator"

Frontend: Text upload interface, tone highlighting, rewrite suggestions

Backend: Cloud Function analyzing text for passive-aggressive tones, suggesting rewrites

Database: textTone/{sessionId} with original texts, analyses, rewrites

AI Integration: NLP analysis for tone detection, GPT-4 for rewrite suggestions

Game 15: 🏆 "Repair Relay"

Frontend: Conflict description display, rapid repair typing interface, timer

Backend: Cloud Function ranking repair attempts by empathy and feasibility

Database: repairRelay/{sessionId} with conflicts, repair attempts, rankings

AI Integration: GPT-4 evaluating repair attempts for emotional intelligence

Game 16: 🎵 "Soundtrack Sync"

Frontend: Music player interface, song selection, mood matching game

Backend: Cloud Function analyzing song lyrics and tempo for emotional alignment

Database: soundtrackSync/{sessionId} with song choices, analyses, matches
AI Integration: Lyric sentiment analysis, musical mood classification

Game 17: 📸 "Micro-Moment Museum"

Frontend: Photo upload interface, caption input, museum gallery display
Backend: Cloud Function curating photos, analyzing caption warmth and specificity
Database: microMoment/{sessionId} with photos, captions, analysis scores
AI Integration: Image sentiment analysis, caption emotional content evaluation

Game 18: 🧪 "Stress Synergy Lab"

Frontend: Stressor rating sliders, overlap visualization, soothing plan builder
Backend: Cloud Function finding stress overlaps, suggesting mutual soothing strategies
Database: stressSynergy/{sessionId} with stress ratings, overlaps, plans
AI Integration: Pattern detection in stress data, GPT-4 for soothing plan generation

Game 19: 🎯 "Dream Support Sprint"

Frontend: Dream input interface, support idea generation, feasibility rating
Backend: Cloud Function evaluating support ideas for concreteness and feasibility
Database: dreamSupport/{sessionId} with dreams, support ideas, ratings
AI Integration: GPT-4 analyzing dream-support compatibility, suggesting improvements

Game 20: 🏃 "Turning Toward Tally"

Frontend: Bid notification system, response timer, streak tracking dashboard
Backend: Cloud Function monitoring bid-response timing, calculating attunement scores
Database: turningToward/{sessionId} with bids, responses, timing data, streaks
AI Integration: Pattern analysis in bid-response timing, streak detection

Game 21: 🎲 "Commitment Dice"

Frontend: Dice roll animation, commitment prompt display, completion timer

Backend: Cloud Function generating commitment prompts, validating completion

Database: commitmentDice/{sessionId} with prompts, completions, vulnerability scores

AI Integration: GPT-4 generating meaningful commitment prompts based on relationship stage

Game 22: 🗣️ "Empathy Echo"

Frontend: Worry recording interface, validation recording, blind rating system

Backend: Cloud Function analyzing validation responses for empathy vs fixing

Database: empathyEcho/{sessionId} with worries, validations, empathy scores

AI Integration: Voice sentiment analysis, empathy vs problem-solving detection

Game 23: 🧩 "Compromise Jenga"

Frontend: Virtual Jenga tower, compromise block addition, stability simulation

Backend: Cloud Function simulating tower stability, evaluating compromise balance

Database: compromiseJenga/{sessionId} with compromises, tower states, stability scores

AI Integration: Physics simulation, compromise equity analysis

Game 24: 📅 "Ritual Roulette"

Frontend: Spinning wheel animation, ritual generation, completion tracking

Backend: Cloud Function generating ritual combinations, tracking completion proof

Database: ritualRoulette/{sessionId} with rituals generated, completions, proof

AI Integration: GPT-4 generating creative ritual combinations

Game 25: 🤖 "Role Swap"

Frontend: Conflict description input, role-swapped text replay, perspective rating

Backend: Cloud Function analyzing perspective-taking accuracy in role swaps

Database: roleSwap/{sessionId} with conflicts, role swaps, accuracy scores
AI Integration: GPT-4 evaluating perspective-taking in rewritten conflicts

Game 26: 🧠 "Memory Lane Dash"

Frontend: Rapid-fire memory recall, timer, accuracy scoring
Backend: Cloud Function pulling old journal entries, validating recall accuracy
Database: memoryDash/{sessionId} with entries shown, recalls, accuracy scores
AI Integration: Semantic matching of memory recall to original entries

Game 27: 🎯 "Admiration Aim"

Frontend: AR target practice interface, word shooting, accuracy scoring
Backend: Cloud Function tracking shot accuracy, distinguishing strengths from criticisms
Database: admirationAim/{sessionId} with words shot, accuracy, patterns
AI Integration: Word classification (strength vs criticism), pattern analysis

Game 28: 📜 "Vow Remix"

Frontend: Vow writing interface, modern reality integration, likelihood rating
Backend: Cloud Function analyzing vow updates for relevance and humor
Database: vowRemix/{sessionId} with vows, ratings, humor scores
AI Integration: GPT-4 evaluating vow creativity and contemporary relevance

Game 29: 🎲 "Legacy Dice"

Frontend: Dice roll, legacy prompt display, voice recording interface
Backend: Cloud Function generating legacy prompts, analyzing voice clarity and warmth
Database: legacyDice/{sessionId} with prompts, recordings, analysis scores
AI Integration: Voice sentiment analysis, legacy value extraction

Game 30: 🌟 "Marcie's Grand Finale: The Connection Conundrum"

Frontend: 10-round rapid-fire interface mixing all game types, progress tracker
Backend: Cloud Function orchestrating mixed game types, calculating final scores
Database: connectionConundrum/{sessionId} with round results, final scores,

custom ritual

AI Integration: GPT-4 generating personalized ritual based on performance patterns

CATEGORY 2: 30 INFIDELITY HEALING GAMES

TRUST THERMOMETER GAMES (5 Games)

Game 31: "Lie Detector: Lite™"

Description: Voice response analysis for transparency - detects pauses, filler words, vocal steadiness

Frontend: Voice recording interface with 10-second timer, waveform visualization, analysis display

Backend: Cloud Function analyzing audio for pauses >1.2s, pitch variance, filler word count

Database: lieDetector/{sessionId} with recordings, analysis results, scores

AI Integration: Speech-to-text with pause detection, vocal pattern analysis

Game 32: "Transparency Toss"

Description: Real-time truth sharing with verification - low-stakes truth "tossing" between partners

Frontend: Shared board interface, truth input, verification buttons, 60-second timer

Backend: Cloud Function managing truth sharing queue, verification tracking

Database: transparencyToss/{sessionId} with truths shared, verifications, timing

AI Integration: Truth validation patterns, verification timing analysis

Game 33: "Boundary Bingo"

Description: 4×4 grid of boundaries - mark only after mutual verification

Frontend: Bingo grid interface, boundary squares, mutual verification system

Backend: Cloud Function generating personalized boundary squares, verifying evidence

Database: boundaryBingo/{sessionId} with grid, verifications, evidence, bingo status

AI Integration: GPT-4 generating personalized boundary challenges

Game 34: "The Apology Olympics"

Description: Speed rewrite of ineffective apologies following specific rules

Frontend: Apology display, rewrite timer, rule checklist, scoring display

Backend: Cloud Function analyzing rewrites for ownership, impact naming, repair offers

Database: apologyOlympics/{sessionId} with apologies, rewrites, scores, analysis

AI Integration: NLP analysis for apology components, blame-shifting detection

Game 35: "Vibe Sync"

Description: Emotional battery setting and guessing with real-time sync

Frontend: Hidden slider for setting, guess slider for partner, real-time value display

Backend: Cloud Function syncing slider values, calculating accuracy percentages

Database: vibeSync/{sessionId} with set values, guesses, accuracy scores

AI Integration: Pattern analysis in emotional attunement over time

BETRAYAL AFTERMATH GAMES (5 Games)

Game 36: "Trigger Takedown (AR)"

Description: AR interface to identify and "squash" triggers with grounding techniques

Frontend: AR camera view, trigger detection, grounding technique selection

Backend: Cloud Function identifying triggers via image recognition, suggesting grounding

Database: triggerTakedown/{sessionId} with triggers identified, grounding used, effectiveness

AI Integration: Computer vision for trigger identification, ML for grounding recommendation

Game 37: "Rewrite the Memory"

Description: Collaborative rewriting of painful memories with hope/absurdity

Frontend: Shared text editor, memory fragment display, collaborative editing

Backend: Cloud Function analyzing rewrites for agency, nuance, humor

Database: rewriteMemory/{sessionId} with original fragments, rewrites,

analysis

AI Integration: GPT-4 evaluating narrative shifts from pain to hope/absurdity

Game 38: "Guilt vs. Shame Sort"

Description: Rapid Tinder-style swipe sorting of guilt vs shame statements

Frontend: Card swipe interface, statement display, streak counter, timeout system

Backend: Cloud Function generating guilt/shame statements, validating swipes

Database: guiltShameSort/{sessionId} with statements, swipes, accuracy, streaks

AI Integration: Statement classification, pattern recognition in sorting mistakes

Game 39: "Flashback Frenzy"

Description: Image association game linking triggers to memories/feelings

Frontend: Image display, emotion word input, partner guessing interface

Backend: Cloud Function analyzing associations, validating memory matches

Database: flashbackFrenzy/{sessionId} with images, associations, guesses, matches

AI Integration: Image-emotion association analysis, memory validation

Game 40: "The Denial Detector"

Description: Voice analysis detecting overuse of "fine" and avoidance language

Frontend: Voice recording prompt, word frequency display, denial scoring

Backend: Cloud Function analyzing speech for avoidance patterns, "fine" frequency

Database: denialDetector/{sessionId} with recordings, word counts, denial scores

AI Integration: Speech pattern analysis, avoidance language detection



CONNECTION LAB GAMES (5 Games)

Game 41: "Six-Second Stare-Down"

Description: Front camera eye contact challenge with AI gaze detection

Frontend: Dual camera view, gaze tracking overlay, 6-second timer

Backend: Cloud Function processing camera feeds for mutual gaze detection
Database: sixSecondStare/{sessionId} with gaze data, duration, sync scores
AI Integration: Face mesh analysis for eye contact detection, blink synchronization

Game 42: "Gratitude Graffiti"

Description: 60-second rapid appreciation writing generating metaphorical word cloud

Frontend: Rapid text input, real-time word cloud visualization, metaphor requirement

Backend: Cloud Function generating word cloud, detecting metaphors, analyzing novelty

Database: gratitudeGraffiti/{sessionId} with words, metaphors, novelty scores

AI Integration: Metaphor detection, word novelty analysis

Game 43: "Vulnerability Volley"

Description: Timed vulnerability sharing with validation ping-pong

Frontend: Prompt display, 15-second response timer, validation input

Backend: Cloud Function timing responses, detecting vulnerability words, validating phrases

Database: vulnerabilityVolley/{sessionId} with prompts, responses, validations, scores

AI Integration: Vulnerability language detection, validation phrase analysis

Game 44: "The Touch Map: Lite"

Description: Abstract body outline for consent and preference mapping

Frontend: SVG body outline, color tap zones, comparison overlay

Backend: Cloud Function analyzing touch preferences, identifying mismatches

Database: touchMapLite/{sessionId} with zones, colors, mismatches, resolutions

AI Integration: Pattern analysis in touch preferences, mismatch resolution suggestions

Game 45: "Memory Lane GPS"

Description: Map pin dropping for significant memories with photo proof

Frontend: Interactive map, pin dropping, photo upload with timestamp verification

Backend: Cloud Function verifying photo timestamps, calculating pin proximity

Database: memoryLaneGPS/{sessionId} with pins, photos, distances, confirmations

AI Integration: EXIF data analysis, location validation

PATTERN PATROL GAMES (5 Games)

Game 46: "Avoidance Arcade"

Description: Whac-A-Mole style game hitting avoidance phrases

Frontend: Mole animation interface, avoidance phrase moles, "Say It Now" hammer

Backend: Cloud Function generating avoidance phrases, tracking hit accuracy

Database: avoidanceArcade/{sessionId} with phrases, hits, misses, scores

AI Integration: Phrase generation for different avoidance patterns

Game 47: "The Needs Decoder"

Description: Emoji cipher game guessing unmet needs from partner's emojis

Frontend: Emoji display, need guessing input, validation system

Backend: Cloud Function generating emoji-need mappings, validating guesses

Database: needsDecoder/{sessionId} with emojis, guesses, validations, scores

AI Integration: Emoji-need association analysis, guess validation

Game 48: "Escapism Escape Room"

Description: 60-second puzzle escaping "Netflix Binge Basement" using real-life insights

Frontend: Escape room puzzle interface, clue system, timer

Backend: Cloud Function generating personalized escape puzzles, validating solutions

Database: escapismEscape/{sessionId} with puzzles, clues used, escape time

AI Integration: Personalized puzzle generation based on couple's escapism patterns

Game 49: "The Blame Flip"

Description: Drag-and-drop rewriting of blaming statements into I-statements

Frontend: Blaming statement display, drag words to rewrite, timer
Backend: Cloud Function validating I-statement structure, speed scoring
Database: blameFlip/{sessionId} with statements, rewrites, times, scores
AI Integration: Syntax validation, blame-to-I-statement transformation analysis

Game 50: "Micro-Betrayal Mini-Golf"

Description: Physics-based mini-golf navigating small breaches to repair
Frontend: Mini-golf physics engine, ball control, hole navigation
Backend: Cloud Function managing physics simulation, tracking strokes
Database: microBetrayalGolf/{sessionId} with breaches, strokes, repairs, scores
AI Integration: Physics simulation, repair path optimization analysis



PLAYFUL TENSION GAMES (5 Games)

Game 51: "The 6-Second Kiss Timer"

Description: Dual thumb-hold pressure synchronization for 6 seconds
Frontend: Pressure-sensitive touch interface, progress bar, synchronization display
Backend: Cloud Function processing touch pressure data, detecting synchronization
Database: sixSecondKiss/{sessionId} with pressure data, sync scores, durations
AI Integration: Pressure pattern analysis, synchronization detection

Game 52: "Foreplay Forecast"

Description: Hidden arousal slider with real-time display for non-verbal communication
Frontend: Hidden slider controller, real-time value display, action suggestion interface
Backend: Cloud Function syncing slider values in real-time, suggesting actions
Database: foreplayForecast/{sessionId} with slider values, actions, effectiveness
AI Integration: Arousal pattern prediction, action recommendation

Game 53: "Ransom Note Romance"

Description: Drag-and-drop cutout words building absurdly sweet "threats"

Frontend: Word bank, drag-and-drop assembly area, absurdity scoring

Backend: Cloud Function generating word banks, evaluating message absurdity/sweetness

Database: ransomNote/{sessionId} with messages, absurdity scores, partner reactions

AI Integration: Creativity evaluation, humor detection in constructed messages

Game 54: "GIF the Feels"

Description: Giphy integration for finding perfect reaction GIFs to relationship prompts

Frontend: Giphy search interface, side-by-side GIF display, voting system

Backend: Cloud Function integrating Giphy API, judging GIF relevance and humor

Database: gifTheFeels/{sessionId} with prompts, GIFs, votes, winners

AI Integration: GIF relevance analysis, humor scoring

Game 55: "Karaoke Confessional"

Description: Rewritten song lyrics about relationship issues for duet singing

Frontend: Lyrics display with highlighting, voice recording, duet synchronization

Backend: Cloud Function rewriting lyrics, analyzing vocal performance and emotional commitment

Database: karaokeConfessional/{sessionId} with lyrics, recordings, analysis, scores

AI Integration: Lyric rewriting, vocal emotion analysis, duet synchronization detection

FUTURE-FOCUSED GAMES (5 Games)

Game 56: "Date Night Roulette"

Description: Filter-based wheel spin generating unique date ideas

Frontend: Filter selection interface, spinning wheel animation, date idea display

Backend: Cloud Function generating date ideas based on filters, tracking

completions

Database: dateNightRoulette/{sessionId} with filters, spins, ideas, completions

AI Integration: Personalized date idea generation based on couple preferences

Game 57: "The Repair Report Card"

Description: Weekly slider survey rating 5 relationship areas with trend analysis

Frontend: Slider interface for 5 areas, trend visualization, improvement tracking

Backend: Cloud Function analyzing weekly trends, calculating improvement scores

Database: repairReportCard/{sessionId} with weekly ratings, trends, improvements

AI Integration: Trend pattern analysis, improvement prediction

Game 58: "Healing Bingo"

Description: Weekly 5×5 grid of healing actions with verification system

Frontend: Bingo grid interface, action verification, streak tracking

Backend: Cloud Function generating personalized healing actions, verifying completions

Database: healingBingo/{sessionId} with grid, verifications, streaks, bingos

AI Integration: Personalized healing action generation based on couple's needs

Game 59: "The Iceberg Dive"

Description: Nested dropdown exploration from surface emotions to core needs

Frontend: Hierarchical dropdown interface, iceberg visualization, progress tracking

Backend: Cloud Function managing emotion-need hierarchy, validating core need identification

Database: icebergDive/{sessionId} with emotions explored, paths taken, core needs

AI Integration: Emotion-need hierarchy analysis, core need validation

Game 60: "The Commitment Countdown"

Description: 30-day streak of daily micro-actions with achievement unlocking

Frontend: Streak calendar interface, daily action display, achievement unlocks

Backend: Cloud Function tracking daily actions, managing streaks, awarding achievements

Database: commitmentCountdown/{sessionId} with daily actions, streaks, achievements

AI Integration: Pattern analysis in commitment consistency, achievement triggering

FIREBASE IMPLEMENTATION ARCHITECTURE

Database Schema for All 60 Games:

javascript

// Firestore Collections Structure

```
{
  // Core Collections
  couples: {
    coupleId: {
      // Core couple data
    }
  },

  users: {
    userId: {
      // User profiles
    }
  },

  gameSessions: {
    sessionId: {
      // Universal game session structure
      gameType: 'bid_radar',
      coupleId: 'couple_123',
      status: 'active',
      // ... other fields
    }
  }
}
```



```
}  
},
```

```
// Game-specific collections (60 total)
```

```
bidsRadarData: {},  
gentleStartupData: {},  
loveMapData: {},  
antidoteData: {},  
mirrorModeData: {},  
dreamDecoderData: {},  
toneShiftData: {},  
ritualBuilderData: {},  
conflictDiceData: {},  
appreciationData: {},  
floodingData: {},  
loveMapGapData: {},  
sharedMeaningData: {},  
textToneData: {},  
repairRelayData: {},  
soundtrackData: {},  
microMomentData: {},  
stressSynergyData: {},  
dreamSupportData: {},  
turningTowardData: {},  
commitmentData: {},  
empathyData: {},  
compromiseData: {},  
ritualRouletteData: {},  
roleSwapData: {},  
memoryDashData: {},  
admirationData: {},  
vowData: {},  
legacyData: {},  
connectionConundrumData: {},
```

```
// Infidelity Healing Games
```

```
lieDetectorData: {},  
transparencyTossData: {},
```

```
boundaryBingoData: {},
apologyOlympicsData: {},
vibeSyncData: {},
triggerTakedownData: {},
rewriteMemoryData: {},
guiltShameSortData: {},
flashbackFrenzyData: {},
denialDetectorData: {},
sixSecondStareData: {},
gratitudeGraffitiData: {},
vulnerabilityVolleyData: {},
touchMapLiteData: {},
memoryLaneGPSData: {},
avoidanceArcadeData: {},
needsDecoderData: {},
escapismEscapeData: {},
blameFlipData: {},
microBetrayalData: {},
sixSecondKissData: {},
foreplayForecastData: {},
ransomNoteData: {},
gifTheFeelsData: {},
karaokeConfessionalData: {},
dateNightRouletteData: {},
repairReportCardData: {},
healingBingoData: {},
icebergDiveData: {},
commitmentCountdownData: {},
```

```
// Supporting Collections
```

```
aiAnalyses: {},
voiceRecordings: {},
mediaStorage: {},
achievements: {},
streaks: {},
leaderboards: {},
analytics: {}
```

```
}
```

Cloud Functions Structure:

typescript

// Functions directory structure for 60 games

/functions/src

```
├── games/
│   ├── gottman/
│   │   ├── bidRadar.function.ts
│   │   ├── gentleStartup.function.ts
│   │   ├── loveMapSpeedrun.function.ts
│   │   ├── antidoteArena.function.ts
│   │   ├── mirrorMode.function.ts
│   │   ├── dreamDecoder.function.ts
│   │   ├── toneShift.function.ts
│   │   ├── ritualBuilder.function.ts
│   │   ├── conflictDice.function.ts
│   │   ├── appreciationAuction.function.ts
│   │   ├── floodingForecast.function.ts
│   │   ├── loveMapGap.function.ts
│   │   ├── sharedMeaning.function.ts
│   │   ├── textTone.function.ts
│   │   ├── repairRelay.function.ts
│   │   ├── soundtrackSync.function.ts
│   │   ├── microMoment.function.ts
│   │   ├── stressSynergy.function.ts
│   │   ├── dreamSupport.function.ts
│   │   ├── turningToward.function.ts
│   │   ├── commitmentDice.function.ts
│   │   ├── empathyEcho.function.ts
│   │   ├── compromiseJenga.function.ts
│   │   ├── ritualRoulette.function.ts
│   │   ├── roleSwap.function.ts
│   │   ├── memoryDash.function.ts
│   │   ├── admirationAim.function.ts
│   │   ├── vowRemix.function.ts
│   │   ├── legacyDice.function.ts
│   │   └── connectionConundrum.function.ts
│   ├── infidelityHealing/
│   └── trustThermometer/
```

- lieDetector.function.ts
- transparencyToss.function.ts
- boundaryBingo.function.ts
- apologyOlympics.function.ts
- vibeSync.function.ts
- betrayalAftermath/
 - triggerTakedown.function.ts
 - rewriteMemory.function.ts
 - guiltShameSort.function.ts
 - flashbackFrenzy.function.ts
 - denialDetector.function.ts
- connectionLab/
 - sixSecondStare.function.ts
 - gratitudeGraffiti.function.ts
 - vulnerabilityVolley.function.ts
 - touchMapLite.function.ts
 - memoryLaneGPS.function.ts
- patternPatrol/
 - avoidanceArcade.function.ts
 - needsDecoder.function.ts
 - escapismEscape.function.ts
 - blameFlip.function.ts
 - microBetrayal.function.ts
- playfulTension/
 - sixSecondKiss.function.ts
 - foreplayForecast.function.ts
 - ransomNote.function.ts
 - gifTheFeels.function.ts
 - karaokeConfessional.function.ts
- futureFocused/
 - dateNightRoulette.function.ts
 - repairReportCard.function.ts
 - healingBingo.function.ts
 - icebergDive.function.ts
 - commitmentCountdown.function.ts
- ai/
 - marcieAI.function.ts
 - openai.function.ts

```
| |— claude.function.ts
| |— voiceAnalysis.function.ts
| |— textAnalysis.function.ts
| |— imageAnalysis.function.ts
| |— sentiment.function.ts
|— scoring/
| |— calculateScore.function.ts
| |— updateMetrics.function.ts
| |— awardAchievements.function.ts
| |— leaderboard.function.ts
|— realtime/
| |— syncHandler.function.ts
| |— notificationHandler.function.ts
| |— presenceHandler.function.ts
|— media/
| |— audioProcessing.function.ts
| |— videoProcessing.function.ts
| |— imageProcessing.function.ts
| |— compression.function.ts
|— utils/
| |— validation.function.ts
| |— security.function.ts
| |— errorHandling.function.ts
| |— logging.function.ts
```

React Native Components Structure:

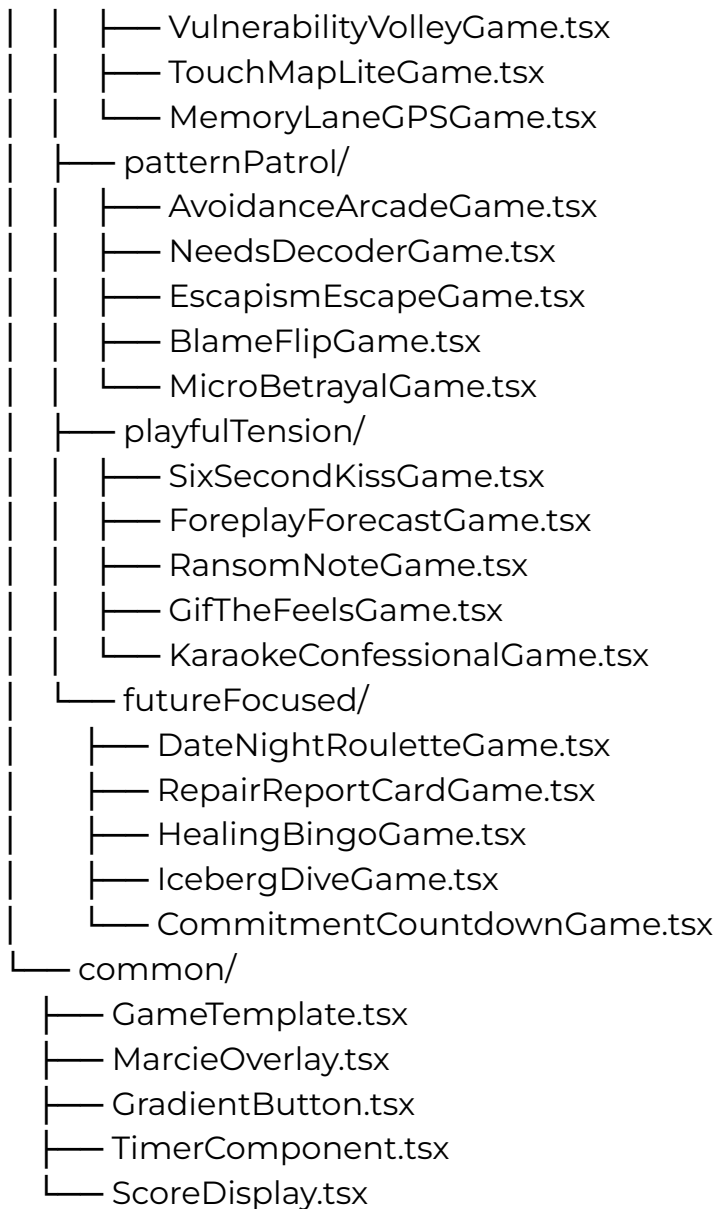
typescript

// React Native components for 60 games

/src/components/games

```
|— gottman/
| |— BidRadarGame.tsx
| |— GentleStartupGame.tsx
| |— LoveMapSpeedrun.tsx
| |— AntidoteArenaGame.tsx
| |— MirrorModeGame.tsx
| |— DreamDecoderGame.tsx
| |— ToneShiftGame.tsx
| |— RitualBuilderGame.tsx
```

- |— ConflictDiceGame.tsx
- |— AppreciationAuctionGame.tsx
- |— FloodingForecastGame.tsx
- |— LoveMapGapGame.tsx
- |— SharedMeaningGame.tsx
- |— TextToneGame.tsx
- |— RepairRelayGame.tsx
- |— SoundtrackSyncGame.tsx
- |— MicroMomentGame.tsx
- |— StressSynergyGame.tsx
- |— DreamSupportGame.tsx
- |— TurningTowardGame.tsx
- |— CommitmentDiceGame.tsx
- |— EmpathyEchoGame.tsx
- |— CompromiseJengaGame.tsx
- |— RitualRouletteGame.tsx
- |— RoleSwapGame.tsx
- |— MemoryDashGame.tsx
- |— AdmirationAimGame.tsx
- |— VowRemixGame.tsx
- |— LegacyDiceGame.tsx
- |— ConnectionConundrumGame.tsx
- |— infidelityHealing/
 - |— trustThermometer/
 - |— LieDetectorGame.tsx
 - |— TransparencyTossGame.tsx
 - |— BoundaryBingoGame.tsx
 - |— ApologyOlympicsGame.tsx
 - |— VibeSyncGame.tsx
 - |— betrayalAftermath/
 - |— TriggerTakedownGame.tsx
 - |— RewriteMemoryGame.tsx
 - |— GuiltShameSortGame.tsx
 - |— FlashbackFrenzyGame.tsx
 - |— DenialDetectorGame.tsx
- |— connectionLab/
 - |— SixSecondStareGame.tsx
 - |— GratitudeGraffitiGame.tsx



Key Backend Services for All Games:

typescript

// Core services needed for all 60 games

interface GameEngineServices {

// 1. AI Integration Service

marcieAI: {

generateCommentary(context: GameContext): Promise<string>;

analyzeText(input: string, analysisType: string): Promise<AnalysisResult>;

analyzeAudio(audioBuffer: Buffer): Promise<AudioAnalysis>;

analyzeImage(imageBuffer: Buffer): Promise<ImageAnalysis>;

```

    generateContent(prompt: string, context: any): Promise<any>;
};

// 2. Real-time Sync Service
realtimeSync: {
    subscribeToChannel(coupleId: string, channel: string):
Promise<WebSocket>;
    sendUpdate(channel: string, data: any): Promise<void>;
    getPartnerStatus(coupleId: string, userId: string): Promise<PartnerStatus>;
    syncGameState(gameSessionId: string, state: any): Promise<void>;
};

// 3. Scoring Service
scoringEngine: {
    calculateGameScore(gameType: string, data: any): Promise<ScoreResult>;
    updateCoupleMetrics(coupleId: string, gameResults: any): Promise<void>;
    awardAchievements(coupleId: string, criteria: any): Promise<string[]>;
    updateLeaderboards(gameType: string, scores: any): Promise<void>;
};

// 4. Media Processing Service
mediaProcessor: {
    processAudioRecording(audioFile: Buffer): Promise<ProcessedAudio>;
    processVideoRecording(videoFile: Buffer): Promise<ProcessedVideo>;
    processImageUpload(imageFile: Buffer): Promise<ProcessedImage>;
    compressMedia(file: Buffer, type: string): Promise<Buffer>;
};

// 5. Game Session Management
sessionManager: {
    createGameSession(coupleId: string, gameType: string):
Promise<GameSession>;
    updateGameState(sessionId: string, updates: any): Promise<void>;
    endGameSession(sessionId: string, results: any): Promise<void>;
    getActiveSessions(coupleId: string): Promise<GameSession[]>;
};
}

```


Firestore Configuration for 60 Games:

```
javascript
// firebase.json with all necessary configurations
{
  "firebase": {
    "rules": "firebase.rules",
    "indexes": "firebase.indexes.json"
  },
  "database": {
    "rules": "database.rules.json"
  },
  "hosting": {
    "public": "build",
    "ignore": ["firebase.json", "**/.*", "**/node_modules/**"],
    "rewrites": [
      {
        "source": "**",
        "destination": "/index.html"
      }
    ]
  },
  "functions": {
    "source": "functions",
    "runtime": "nodejs18",
    "timeout": "540s",
    "memory": "2GB",
    "env": {
      "OPENAI_API_KEY": "@openai-api-key",
      "ELEVENLABS_API_KEY": "@elevenlabs-api-key",
      "GIPHY_API_KEY": "@giphy-api-key",
      "GOOGLE_MAPS_API_KEY": "@google-maps-api-key"
    }
  },
  "storage": {
    "rules": "storage.rules"
  }
}
```

Security Rules for All Collections:

```
javascript
// firestore.rules covering all 60 game collections
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {
    // Allow users to read their own data
    match /users/{userId} {
      allow read, write: if request.auth != null && request.auth.uid == userId;
    }

    // Couple data - both partners can read/write
    match /couples/{coupleId} {
      allow read: if request.auth != null &&
        (resource.data.player1Id == request.auth.uid ||
         resource.data.player2Id == request.auth.uid);
      allow write: if request.auth != null &&
        (resource.data.player1Id == request.auth.uid ||
         resource.data.player2Id == request.auth.uid);
    }

    // Game sessions - participants can access
    match /gameSessions/{sessionId} {
      allow read, write: if request.auth != null &&
        request.auth.uid in resource.data.players;
    }

    // All game-specific collections (pattern for 60 games)
    match /[gameCollection]/{documentId} {
      // Generic rule for all game data collections
      allow read: if request.auth != null &&
        request.auth.uid in
get(/databases/{database}/documents/couples/{resource.data.coupleId}).dat
a.players;
      allow write: if request.auth != null &&
        request.auth.uid == resource.data.playerId;
    }
  }
}
```

}

IMPLEMENTATION PRIORITY & TIMELINE

Phase 1: Core Infrastructure

- Firebase project setup with all collections
- Authentication system with couple linking
- Dr. Marcie AI integration with ElevenLabs
- Base game template component
- Real-time synchronization system

Phase 2: Gottman Games Core

- Games 1-10: Bid Radar through Appreciation Auction
- Core AI analysis functions
- Basic scoring and achievement system
- Real-time multiplayer sync

Phase 3: Gottman Games Advanced

- Games 11-20: Flooding Forecast through Turning Toward Tally
- Media processing (audio/video/image)
- Advanced AI integrations
- Achievement and badge system

Phase 4: Gottman Games Complete

- Games 21-30: Commitment Dice through Connection Conundrum
- All AI features complete
- Comprehensive scoring system
- Leaderboards and social features

Phase 5: Infidelity Healing Games Part 1

- Games 31-40: Trust Thermometer through Denial Detector
- Specialized healing game mechanics
- Privacy and sensitivity features
- Trauma-informed design

Phase 6: Infidelity Healing Games Part 2

- Games 41-50: Connection Lab through Pattern Patrol
- AR and camera features
- Advanced media processing
- Healing progress tracking

Phase 7: Infidelity Healing Games Completed

- Games 51-60: Playful Tension through Future Focused
- All game mechanics complete
- Comprehensive analytics
- Offline mode support

Phase 8: Polish & Launch

- Performance optimization
- Security audit
- App store submission
- Marketing site integration